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Diverse perspectives and experiences of university mathematics teachers on improving the secondary-tertiary transition

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Abstract
This study explores the diversity of university mathematics teachers' perspectives and experiences in relation to the secondary-tertiary transition (STT). Data for this study consist of responses of 310 university mathematics teachers from 30 countries to a survey. The survey design and data analysis are informed by an adaptation of a theory of improvement for education and by four idealized approaches for improving STT as identified in the literature: jump-oriented, enculturation-oriented, diversity-oriented, and cross-community-oriented. We characterize the diversity of perspectives and experiences in the data and examine ideas for future projects in light of these four approaches. We conclude that while jump-oriented approaches for improving STT are still the most prevalent, in part due to a persistent perspective of STT as a "deficit," university mathematics teachers' discourse on STT is becoming more diverse and nuanced. We find it particularly promising that participants' suggestions for future action are aligned with the four approaches, and most notably, with the cross-community approach expressing the emergent need for greater communication between the secondary and tertiary mathematics education communities.

Keywords Secondary-tertiary transition in mathematics · Teachers' perspectives · Improvement science · Deficit narratives · International survey

1 Introduction

Students' difficulties during their transition from studying secondary-level mathematics to studying tertiary-level mathematics—a process known as the *secondary-tertiary transition* or STT—have been receiving keen research attention for more than 3 decades, particularly in light of growing concerns regarding enrollment numbers and dropout rates in tertiary STEM programs (Hernandez-Martinez, 2016; PCAST, 2012; Rach & Heinze, 2017). In addition to the substantial emotional stress for the involved students (Di Martino &

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Gregorio, 2019), research has shown that STT has considerable social and economic implications (Biza et al., 2022; Ellis et al., 2014; Good et al., 2008; Thomas et al., 2015).

The discourse about STT has changed considerably over the decades. Early work in this area, mostly addressing what was referred to at the time as "advanced mathematical thinking," has focused on epistemological differences between mathematics as practiced at school and as practiced by mathematicians and on the individual cognitive processes underlying the transition between these two kinds of mathematics (e.g., Tall, 1991). From the mid-1990s, attention has gradually shifted from the individual student towards sociocultural, institutional, and affective aspects of STT that have substantial impact on student learning (Artigue et al., 2007; Clark & Lovric, 2009; De Guzmán et al., 1998; Di Martino & Gregorio, 2019; Gueudet, 2008; Selden, 2012). Along with these developments, the perception of STT has been revisited, at least in mathematics education research if not in practice. For many years, STT was studied as a process that entails not only learning new mathematics, but also unlearning the old (school) mathematics. However, over time, more and more studies have been relating to STT less as redemption and more as enculturation and accommodation of new "rules of the game," without necessarily taking a normative stance about which set of rules is superior (Dawkins & Weber, 2017; Niss, 2003; Sfard, 2014; Watson, 2008).

In parallel to the developments in research, and to increasing public accountability and scrutiny of instructional quality and student (dis)satisfaction in post-secondary institutions (Artigue, 2016), there have been substantial developments in the field as well. Measures for supporting students through STT were called for, suggested, and tried in practice (Bardelle & Di Martino, 2012; Cronin & Meehan, 2021; Hochmuth et al., 2021; Matthews et al., 2013; Thomas et al., 2015; Wood, 2001). However, as is the case in other areas of mathematics education, the relationship between the developments in research and in practice is complex. Whereas at least some of the initiatives for improving the STT are accompanied by research, our knowledge of the effectiveness, impact, and sustainability of current efforts to address STT is scarce and fragmented (Hochmuth et al., 2021). From the little we know, it appears that many if not most initiatives for improving the STT are confined to a single mathematics department and are not informed through communication between different tertiary institutions or between practitioners and researchers. As a result, researchers, practitioners, and policy-makers who wish to improve the situation with respect to STT have little access to knowledge accumulated in the field regarding the implementation of different types of measures for addressing STT or regarding how university mathematics teachers (UMTs henceforth) engage with them.

Recognizing that these lacunas in knowledge must be filled in order to devise an informed agenda for further action, the Education Committee of the European Mathematics Society initiated a survey of UMTs' perspectives and experiences on STT and on their ideas regarding further action (Koichu & Pinto, 2019). For many years, research on STT mostly focused on the mathematics and on students' difficulties, and only little attention was given to the perspectives and attitudes of UMTs towards STT. In focusing on UMTs, we join other researchers (Corriveau, 2017; Thomas & Klymchuk, 2012; Thomas et al., 2015) who recognized the importance of listening to the voices of those coping with STT on a day-to-day basis over many years.

The survey received 310 responses from 30 countries. Most of the respondents were from Europe (200 respondents), the USA (40 respondents), and Australia (29 respondents). Admittedly, the survey sample was too small to be representative of the complex STT phenomena. Nevertheless, the responses to the survey provide a rich corpus of perspectives and ideas of UMTs who wanted their voices to be heard and to be taken into account.

Accordingly, we argue that the survey is a good source of information about the diverse perspectives and ideas regarding STT among UMTs who take an active stance towards improving STT and are either involved, have been involved, or look forward to be involved in efforts to improve STT. Moreover, we argue that the survey can provide valuable information about ways in which research on STT have informed or could inform the field. Accordingly, in this study, we aim to explore the perspectives and experiences of UMTs in relation to improving STT and how these perspectives and experiences compare with approaches suggested in the literature for improving STT.

This paper is structured as follows. In Section 2, we review literature on STT and delineate four idealized approaches for improving STT. We then review literature on UMTs' perspectives on STT and improve it and articulate our research question. In Section 3, we examine STT from the perspective of improvement science in order to operationalize our research question and clarify what we mean by UMTs' perspectives on improving STT. In Section 4, we describe our methodology, and in Section 5, we present our findings, which are discussed in Section 6.

2 Theoretical background

2.1 Different approaches for improving STT in the scholarly literature

As mentioned, early research related to STT aimed to understand the epistemic differences and cognitive processes underlying students' difficulties in their initial encounters with tertiary mathematics (De Guzmán et al., 1998; Edwards, et al., 2005; Tall, 1991). Often, these difficulties were painted as stemming from deficiencies in secondary mathematics education, as illustrated in the following description of De Guzmán et al. (1998):

> Secondary school students often succeed in mathematics by relying on their ability to perform algorithms and in spite of a lack of a real understanding of the mathematical concepts with which they are working; they may then experience substantial difficulties, when moving to the tertiary level, in being able to participate by themselves in the process of mathematical thinking, and not merely learn to reproduce mathematical information. (p. 752)

Similarly, other works that attended to STT have addressed students' unpreparedness for tertiary mathematics while describing secondary mathematics education as "surface learning" (Selden, 2005) or a superficial encounter with mathematics (Artigue, 2004). Notably, secondary mathematics education is often portrayed as not only falling short in terms of *what* mathematics students learn, but also, and perhaps even more so, in *how* students learn the mathematics and which understandings they develop (Gueudet, 2008; Clark & Lovric, 2009). This perspective is based in part on the observation that STT-related difficulties are not limited to a portion of students who are less well prepared, but also to high-attaining students, who still appear to be deficient in fundamental aspects of learning mathematics (Biza et al., 2016; Thomas et al., 2015). Two common approaches for improving STT that are aligned with that perspective are to bring secondary mathematics curricula closer to the discipline (e.g., Cuoco et al., 1996; Wu, 2011) or to propose bridging courses that help students close the secondary-tertiary gap (Wood, 2001). If STT is viewed as a case of a discontinuity in mathematics education (Gueudet et al., 2016), then such approaches imply that STT is a *jump-discontinuity*. Accordingly, in what follows, we use the term

jump-oriented for approaches for improving STT by attending to deficiencies in students' preparedness for tertiary mathematics before or shortly after tertiary studies commence.

While at the beginning of the millennium research on STT was relatively scarce (Selden & Selden, 2001), in the decade that followed, research on STT started to accumulate (Gueudet, 2008), with more focus being placed on disciplinary values, norms, practices, and other sociocultural and institutional aspects of tertiary mathematics education (Artigue, 2016). Accordingly, discussion on STT has extended to include aspects of the tertiary learning environments that tacitly condition and constrain student learning during STT (Artigue et al., 2007; Clark & Lovric, 2009; Gueudet, 2008; Iannone & Nardi, 2005; Weber, 2004). During the second decade of the millennium, researchers' attention to social and institutional aspects of STT extended further, to include empirical studies of teaching styles and instructional approaches in tertiary mathematics courses and the ways of doing, communicating, and thinking about mathematics that they make available, explicitly or implicitly, for students (Alcock, 2010; Biza et al., 2016; Corriveau, 2017; Di Martino & Gregorio, 2019; Güçler, 2013; Lew et al., 2016; Nardi et al., 2014; Pinto, 2019; Pinto & Karsenty, 2018, 2020; Weber, 2010, 2012). In accordance with the growing recognition of STT as an enculturation process that cannot be fully addressed within short-term measures such as bridging courses, suggestions for improving STT have placed greater emphasis on long-term interaction and communication between students and UMTs, as representatives of the community of tertiary mathematics education (Artigue, 2016; Güçler, 2016; Pinto, 2019; Sfard, 2014). One particular example of such measure that has received much attention and has become relatively popular in several countries is mathematical support centers (Lawson et al., 2022; Matthews et al., 2013; Schürmann et al., 2020). In what follows, we refer to suggestions for improving STT that emphasize prolonged communication and interaction between UMTs and students as *enculturation-oriented*.

A third approach we identified in the literature for improving STT is one that calls for its diversification, namely improving STT by recognizing that different students, even in the same course, may be going through substantially different transitions. Research on tertiary mathematics education is in many ways oriented towards the mathematical education of future pure mathematicians or mathematics teachers and their particular needs (Artigue, 2016; Artigue et al., 2007; Biza et al., 2022). Consequently, the diversity of forms of professional relationship with mathematics for which university courses may prepare graduates is often not sufficiently investigated and taken into account (Artigue, 2016). Accordingly, researchers have been calling for research and pedagogy that address in meaningful ways the diverse social, cultural, and institutional cohorts of tertiary mathematics education (Biza et al., 2022). Another kind of diversification of STT that is gaining increased attention in research, mostly in the USA, is the recognition of the diverse experiences of students based on individual characteristics such as race, gender, sexual orientation, or cultural background (Adiredja & Zandieh, 2020; Ellis et al., 2014; Leyva et al., 2021). We refer to approaches for improving STT that aim to address diversity, equity, and inclusion (DEI) as *DEI-oriented.*

Finally, there are calls from researchers for improving STT through "collaborations between communities of teachers and professionals across the transition divides" (Hochmuth et al., 2021, p. 210). Such collaborations are considered essential for overcoming issues of coherence in mathematics as a discipline in which the rules of the game appear to change markedly across educational levels and professional contexts (Niss, 2003; Sfard, 2014). Indeed, the Education Committee of the European Mathematics Society that initiated this study includes mathematicians, UMTs, and mathematics education researchers, and one of the goals of the Committee was to prepare the grounds for future collaborations

(Koichu & Pinto, 2019). We refer to suggestions for improving STT that are based on forming cross-community partnerships as *cross-community-oriented*.

2.2 Prior research on UMTs' perspectives on STT

Research on UMTs' perspectives on STT and how it could be improved is still relatively scarce. One exception is the study of the ICME12 Survey Team 4 (Thomas et al., 2015), which presented perspectives regarding STT of 79 UMTs from 21 countries. Notably, 56 of these UMTs mentioned at least one academic support measure to assist students in the transition from school to university employed at their mathematics departments. Most of the respondents of Thomas et al.'s survey thought STT ought to be smooth yet did not consider STT to be smooth in their respective departments. Twenty-six respondents attributed students' difficulties during STT to lack of preparation in secondary school. Thomas et al. concluded that:

> Overall, the survey confirmed that students do have some difficulties in transition, and these are occasionally related to a deficit in student preparation or mathematical knowledge. However, there are also a number of areas that universities could address to assist students, such as adjusting the content of first year courses, and instituting a course on proving and proof (where this doesn't already exist) and constructing appropriate bridging courses. (p. 271)

Thomas et al.'s conclusion highlights how suggestions to improve STT often identify or imply some sort of student-related *deficit* that needs to be addressed. In the scholarly literature, deficit narratives have been defined as socially circulated and reified stories that "focus on students' academic and intellectual shortcomings and attribute them to deficiencies located in the students, their values, families, and/or cultures" (Adiredja & Zandieh, 2020, p. 240). We stress that deficit narratives can theoretically be present (or absent) in all four approaches, but especially in jump-oriented and enculturation-oriented ones when normative stances are taken. Deficit narratives have been a central topic of educational research in K-12, where they were found to influence policy and limit learning opportunities in classrooms in a variety of ways, which are almost always amplified for historically marginalized students (Louie et al., 2021). While we could not locate research that attends explicitly both to STT and to deficit narratives, we note that there is an emerging body of work, mostly conducted in the USA, which suggests that deficit narratives may also have a key role in tertiary mathematics education (Adiredja & Zandieh, 2020; Ellis et al., 2014; Leyva et al., 2021; Peck, 2021; Sabbah & Heyd-Metzuyanim, 2021).

Whereas Thomas et al.'s conclusion, quoted above, may give the impression that UMTs' perspectives on STT are generally jump-oriented and include deficit narratives, we note that Thomas et al.'s findings actually highlight diversity between the UMTs, as these orientations represent only about one-third of the UMTs that participated in the study. One is left wondering what the perspectives of the other UMTs were. Moreover, in relation to improving STT, while most responses focused on changes at the secondary level or bridging courses, which are aligned with a jump-oriented approach, many respondents also supported or at least referred to their experiences in relation to various academic measures that are aligned with other approaches for improving STT. All this underlines the need to delineate and document UMTs' perspectives on and experiences of improving STT that are not jump-oriented. In this paper, we look to build on and continue Thomas et al.'s work while being guided by the following overarching research question: What are the diverse

perspectives and experiences of UMTs in relation to improving STT? This question is operationalized in the next section.

3 STT from the perspective of improvement science

As shown, STT is commonly perceived as a process to be improved. The widespread improvement-related discourse on STT makes it a research object that can be explored by means of approaches offered by improvement science in its educational version (Bryk et al., 2015; Lewis, 2015; Peurach et al., 2022). Improvement science suits our research aims of exploring UMTs' potentially evolving perspectives and experiences in relation to STT as it considers improvement from a bottom-up approach that focuses on identification of problems of practice, their analysis, specification, and solution by actively participating in local contexts, using evidence gained from those contexts (Bryk et al., 2015; Cai & Hwang, 2021; LeMahieu et al., 2015; Russell & Penuel, 2022).

In broad terms, improvement science deals with the relationship between improvement and accumulation of knowledge. Perla et al. (2013) assert that *improvement* has meaning only when related to observation based on a given criterion for interpreting and evaluating change (e.g., efficiency, alignment with stated goals, or expectations). In turn, these criteria heavily depend on the past experiences of the observers, UMTs in our case, and therefore are more pragmatic than normative: the criteria focus on what is "useful" from the improvers' perspective rather than on what is "true" or "right." Furthermore, the improvement discourse features interventionalist approach to education where interventions—in our case, specific measures for addressing STT—are tailored to the local contexts and are undertaken in order to address local problems and needs.

Lewis (2015) distinguishes between two kinds of knowledge that can support improvement in an educational system: *basic knowledge* of the discipline and *profound knowledge* about enacting basic knowledge in particular organizations. In the case of STT, basic knowledge attends to the nature of the transition students undergo in general or specifically in the university of the UMT. For example, it attends the different characteristics of doing and understanding mathematics at the tertiary level and at the secondary level or to the different meanings of key mathematics concepts at these levels, as was the focus of Thomas et al. (2015). However, basic knowledge can only partly support mathematics departments looking to improve the situation with respect to STT. Development and systematic implementation of effective and sustainable measures for improving STT necessarily depends on variables of different scope and scale, including institutional norms and the availability of resources. For example, the department can rely on existing (basic) knowledge in identifying how certain student difficulties with respect to key mathematical concepts are related to different meanings they assume in tertiary mathematics (Thomas et al., 2015). However, it is still to be decided *when* and *how* to deal with these well-anticipated difficulties. Would a measure situated in a jump-oriented approach, such as a bridging course, address well these difficulties in the local context? Would the students have time to attend such a course and motivation to engage with these key mathematical concepts in this setting? Or would a measure related to an enculturation-oriented approach, such as individual mentoring, be more effective? Or should resources be allocated to professional development of UMTs? Do UMTs have the time and motivation?

Knowledge pertaining to these questions consists of *generalizable profound knowledge* (e.g., knowledge of how bridging courses can be organized or knowledge of digital

platforms that can support students in STT) and *organization-specific profound knowledge* (e.g., knowledge of the resources available in a university and knowledge of incentives of the main stakeholders). From the perspective of improvement science, Thomas et al.'s (2015) study, which focused on key mathematics concepts in relation to STT, provided valuable insight into UMTs' basic knowledge for improving STT. In our study we aim to complement Thomas et al.'s work by focusing on UMTs' profound knowledge for improving STT, guided by the following six core principles for applying improvement science to the work of improvement in education (Bryk et al., 2015): (P1) making the work of improvement problem specific and user centered; (P2) focusing on variations in performance; (P3) seeing the system that produces current outcomes; (P4) identifying measurable outcomes of improvement at scale; (P5) using disciplined inquiry to drive improvement; and (P6) accelerating learning through networked communities. These principles serve as an organizational framework for collecting and analyzing the data, as we described in Section 4.

Finally, we operationalize our research question building on three fundamental questions that according to Lewis (2015) drive improvement work: (1) What are we trying to accomplish? (2) How will we know that a change is an improvement? (3) What change can we make that will result in improvement? In relation to UMTs' perspectives on and experiences of improving STT, we rephrase Lewis's fundamental questions into the following research questions that operationalize the aforementioned overarching question of our study:

(FQ1) What are UMTs looking to improve with respect to STT?
(FQ2) How do UMTs determine whether there is an improvement?
(FQ3) What changes do UMTs propose for making an improvement?

4 Methodology

Data for this study were collected through a survey initiated by the Education Committee of the European Mathematics Society in 2019. Since we intended to investigate the diverse perspectives in the field and include voices of UMTs that do not interact regularly with mathematics education researchers, we opted to collect responses through an online questionnaire and to use questions that would be sensible to UMTs. An online questionnaire also serves the purpose of giving to the respondents the chance to answer anonymously, which is important for minimizing pressure and potential risks that could create biases (Mitchell & Jolley, 2010). However, those respondents who were interested to establish via the survey a channel of communication with the Committee or consider options for future collaboration were given such a chance.

The questionnaire[1] was distributed via the European Mathematics Society and ICME Newsletters and appropriate national and international mailing lists (Koichu & Pinto, 2019). The survey consisted of three parts and 10 items as summarized in Table 1. The parts and items of the questionnaire were aligned with the three fundamental questions (FQ1-FQ3) and six core principles (P1-P6) discussed in the previous section. Each part started with a closed item, followed by open items for elaborating and eliciting UMTs' perspectives.

[1] A copy of the questionnaire is available as supplementary material.

Table 1 The three parts of the of questionnaire

	Fundamental question	Questionnaire items (I#)
Part A	What are UMTs looking to improve with respect to STT, if at all?	I1. STT is an issue of concern. How do you feel about this statement? (closed question)
		I2. In your experience, in what ways have concerns regarding STT changed over the last 10 years?
Part B	How do UMTs determine whether there is an improvement?	I3. For each of the following measures, please indicate whether, to your knowledge, your institution employs this measure, and to what extent it is effective for supporting students throughout STT: (closed question)
		I4. Other institutional measures?
		I5. Please elaborate on one of the measures employed in your institution and explain its effectiveness or ineffectiveness in supporting students throughout STT
Part C	What changes do UMTs propose for making an improvement?	I6. Considering that resources for addressing STT are limited, how would you prioritize substantial reform in each of the following areas: (closed question)
		I7. Other reform measures?
		I8. For each of the following stakeholders, to what extent do you agree that they are in a position to have substantial positive impact on STT? (closed question)
		I9. If there are other stakeholders, please list them and indicate their potential positive impact
		I10. Can you describe in rough detail a realistic project that, given appropriate support, could make things better in terms of STT?

Part A of the questionnaire aimed at exploring UMTs' perspectives regarding whether and if yes what needs to be improved with respect to STT, while eliciting responses that are problem specific and user centered (P1), and also attentive to variations in performance over the years (P2). Part B aimed at eliciting criteria that UTMs may use to determine whether there is improvement, by directing attention to measures being employed at present (P3) and to their perceived effectiveness (P4). Part C, with an eye towards the future, aimed at exploring UMTs' perspectives regarding specific and feasible changes that may result in an improvement. To this end, we directed attention to the broader context in which STT takes place and to the stakeholders involved (P3), inviting concrete ideas about improving STT, while probing interest and openness towards being part of a networked cross-community effort to improve STT (P6). The survey concluded with background questions about the respondent's country, years of teaching experience, and interest in being involved in further action. As mentioned, identifying information such as the participants' institutions was left optional. Consequently, the results of the survey cannot be interpreted as representative for the spread of measures different countries and/or departments employ to address STT. For this reason, and also because of the sensitivity of the data, the information about countries of residence is removed from the responses quoted in Section 5.

Data analysis was guided by the four idealized approaches for improving STT discussed in the Theoretical Background section. We first asked ourselves, for each item of the questionnaire—open or closed—and for each idealized approach, what a response that is aligned with this idealized approach would look like. Based on our a priori answers, we proposed initial codes for each item to indicate a possible alignment between a response and an idealized

approach. For example, with respect to item 10 (proposing a realistic project for improving the situation), we used codes to signify which stages of STT the project addresses (e.g., during secondary mathematics education, before, or shortly after starting tertiary mathematics education). For each item, we proposed several such codes. We then tested and refined our codes on a sample of 60 responses from five countries. First, we made a randomly sorted list of all the answers to each item and coded them. Secondly, for each respondent, we read thoroughly the answers to the whole questionnaire and, for each part, decided whether and to what extent it was aligned with an idealized approach, as described in Section 2. Finally, we compared the answers of the two stages and refined our codes. We then used the refined codes on an additional sample of responses and again tested our codes. Altogether, we analyzed all the respondents' (310) answers to the closed items (I1, I3, I6, I8) and answers of 162 respondents (52% of 310) to open items (I2, I4, I5, I7, I9, I10). To select the 162 responses, we first scanned all the answers to all the open items and marked those that were informative enough to be coded and then collected those responses that provided informative answers to multiple open items. We note that we attended to relationships between answers to different items (e.g., "the proposed project in item 10 builds on the measure addressed in item 5"). While these relationships helped us make additional sense of the results of the survey, in this paper, we mainly focus on whether and how UMTs' answers in different parts of the questionnaire are aligned with the four idealized approaches.

5 Findings

5.1 FQ1: what are UMTs looking to improve with respect to STT, if at all?

In response to item I1, almost all the respondents (95.1% out of 310) agreed with the statement "STT is an issue of concern," and most respondents (60%) agreed strongly. This conclusive result may not be surprising considering that UMTs were less likely to participate in the survey if they did not consider STT a concern. More notably, in the sample of 162 responses that we analyzed qualitatively, 94 responses to item 2 indicated a change over the last decade, of which 73 (77% of 94) attested that the concerns regarding STT have aggravated over the last decade, and only 5 (5% of 94) attested that the situation with respect to STT has improved. The remaining 16 responses (17% of 94) did not refer to changes with respect to STT but noted that attention and accountability to STT at their respective departments have increased.

In 101 responses (62% of 162), the respondents provided an explanation for the aggravation, and 95 of these explanations addressed, to some extent, students' preparedness, readiness, or maturity. Here, an explanation was identified as aligned with a jump-oriented approach when it implied that either students, student preparedness, or secondary mathematics education did not meet the respondent's expectations. We characterized such explanations as deficit narratives when they referred to shortcomings in relation to students, student preparedness, or secondary mathematics education. Here are a few representative excerpts:

> I think the secondary education mathematics has become more focused on applications and modeling and less focused on why questions. (Jump)
> The mathematical level of the high school students and their autonomy to find resources in order to solve mathematical problems has decreased in the last years.

> Consequently, the starting point is a handicap for the next steps in the university. (Jump & Deficit)
> The challenges are not with the students' ability but perhaps in the way that mathematics is presented at school - that is as a list of things to be memorized with an algorithmic approach to problem solving rather than one that is inquiry and discovery driven. The 'why' of mathematics has been replaced by the 'how'. (Jump & Deficit)

Altogether, we identified 84 (83% of 101) responses as jump-oriented and 67 (66% of 101) responses as including deficit narratives. We note with respect to the latter that the percentage we found (66%) is considerably higher than the percentage (33%) reported by Thomas et al. (2015), although this difference may be partly explained by different operationalizations of a "deficit." Respondents often cited more than one kind of deficiency, for example: students' weak mathematical foundations, poor algebraic skills, poor reasoning, unwillingness to engage with rigorous mathematics, tendency to give up, lack of interest or motivation, poor working attitudes, and dependency. Some deficiencies were linked with specific mathematical content, others with general mathematical skills, understandings of mathematics, or attitudes towards mathematics.

Whereas aggravation with respect to STT was most often attributed to changes at the secondary level, in some cases, aggravation was linked with changes at the tertiary level, most notably changes in admissions or social changes outside tertiary education that had an effect on students entering STEM programs. Altogether, we identified 19 (19% of 101) responses that referred to students' preparedness, readiness, or maturity in the context of the heterogeneity of students taking tertiary mathematics courses nowadays. We characterized such responses as deficit narratives when heterogeneity was described in terms of deficiencies (e.g., weaker students) and identified responses as DEI-oriented when heterogeneity was described with positive or neutral sentiment (e.g., different interests or needs). Accordingly, 10 responses were identified as deficit narratives and 9 as DEI-oriented. Here are a few selected excerpts from DEI-oriented responses:

> STT has become more linked to issues of equity and diversity. Students from underprivileged backgrounds need more support to make this transition.
> [STT has] not changed in a mathematical way that I know of. Changes have had more to do with attracting certain groups of people (e.g., encouraging women to pursue studies in mathematics).
> In [country], as more students attend college/community college, there are more students that are first generation college students. First generation college students are less likely to understand that there will be a difference between high school classes and college classes. And even if they do know there is a difference, they are unaware what those differences are.

Compared to responses that contained deficit narratives, DEI-oriented responses more often addressed specific social or cultural developments in their countries, specific economic considerations of their institutes, or specific placement procedures.

Finally, we briefly address 'outliers' responses. These responses attended to various STT challenges such as anachronistic teaching methods at the tertiary level, universities treating students as customers, decreased work span of students due to "cellphone mentality," or insufficient communication between faculty members and secondary teachers. Some of these responses could have been identified as rare cases of enculturation-oriented or cross-community-oriented approaches, for example:

> I don't think the learning leap has changed much. Institutional responses have

improved and we are now looking at the whole transition experience. At my institution we have brought in mentoring, ice breakers and support - which includes more than just learning support. (Enculturation-oriented)

The idea of making sure secondary students are prepared for college-level math has always been a concern. However, the ideas of what constitute 'being prepared' are constantly changing - among society, among secondary teachers, and among university educators - yet there is very little dialogue among these groups. That means that as each group changes ideas and practices to support 'being prepared' the other groups are not going to be familiar with what is happening in each area of a student's academic life. (Cross-community-oriented)

5.2 FQ2: how do UMTs determine whether there is an improvement?

Items I3 and I4, in which respondents were asked to specify which support measures their institute employs to assist students during STT and how effective they are, provided background for exploring FQ2. Responses to these items are summarized in Table 2 (100% stands for 310). We briefly note that the most common measures are those indicated in the literature (cf. Thomas et al., 2015) and that 94% of the respondents were aware of at least one measure employed in their department, and 79% were aware of at least two measures. These results are significantly higher than the results of Thomas et al. (2015), which seems to suggest an increase over the last decade of actions taken within mathematics departments for addressing STT, or at least of UMTs' awareness of these actions.

We also note the variability in the perceived effectiveness of different measures for coping with STT. This suggests that UMTs have some criteria to determine whether measures effectively improve the situation with respect to STT. Of note is that most of the UMTs declared enough knowledge about measures employed in their department to express an opinion on their effectiveness: only in less than 10% of the cases did the UMTs state that they did not have an opinion.

The UMTs' answers to item 5 suggest several criteria UMTs may use to determine whether a certain measures makes an improvement with respect to STT: (a) how many students are making use of the measure (e.g., students do not show up); (b) students making use of the measure are the ones most needing support; (c) students using the measure display motivation and engagement; (d) the measure addresses certain needs of the students that are not addressed effectively by other measures; (e) dropout or failure rates have improved; (f) students exhibit less difficulties.

We note an interesting turn in the perspectives expressed in part B of the questionnaire in comparison with those expressed in part A. Whereas the assertions in part A were typically jump-oriented and contained deficit narratives, in part B respondents expressed dissatisfaction with measures that are aligned with this orientation (e.g., bridging courses), yet

Table 2 Indications of employed measures and their perceived effectiveness in UMTs' responses

$N=310$	Bridging courses	Pedagogical training	Resources for lecturers	Resources for students	Tutoring/mentoring
Employed	71.3%	48.8%	51.7%	85.3%	70.7%
Highly effective	15.1%	7.2%	20.4%	18.8%	35%
Ineffective	19.2%	37.7%	19%	19.6%	13.3%

generally appreciated tutoring and mentoring arrangements, which are more often enculturation-oriented. There seems to be some relationships between responses to part A and responses to part B. Bridging courses were considered more effective by those respondents whose answers to item I1 were jump-oriented, suggesting that the criteria the UMTs used to determine whether current measures improve STT were linked with their perspectives on the nature of STT.

We conclude by noting the almost complete absence of references in the descriptions of existing measures of collaboration with secondary teachers, or to any influence of experience accumulated in other institutions or countries, or to mathematics education research on STT. While the survey did not explicitly ask about cross-community collaborations, we still consider this an indication that cross-community-oriented approaches for improving STT are rare in the field, though they are provisioned by some of the UMTs, as evident from responses to part C of the questionnaire.

5.3 FQ3: what changes do UMTs propose for making an improvement?

Items I6–I10 addressed future actions. Item 6 asked participants to prioritize reforms in various areas pertinent to STT, each area appearing once at the school level and once at the tertiary level (see the questionnaire for details). In the 163 (53% of 310) responses that prioritized exactly one or two of the reforms as "of the highest importance," there was a clear leaning towards school-level reform over tertiary-level reform, with every kind of school reform gaining more "of the highest importance" votes than any kind of tertiary level reform. Of the different areas of reform, teacher training and professional development was the most favorable but only by a slight edge. Notably, 163 UMTs still indicated that reforms in various areas of tertiary level are necessary. In particular, 88% of UMTs indicated reform in lecturer professional development as either important or of the highest importance. This result seems well aligned with the low opinion of many UMTs regarding current tertiary pedagogical training measures (see Table 2).

With respect to stakeholders in a position to have substantial positive impact on STT (item I8, see Table 3), UMTs and school mathematics teachers were indicated the most. Notably, with respect to all other stakeholders, results were far less appreciative, particularly in the case of tertiary students, designers of learning resources, and mathematics education researchers. We note that responses characterized as deficit narratives were significantly less supportive of UMTs being in position to impact STT positively and significantly more supportive of teacher educators and school curriculum and reform committees.

Within the 162 responses we analyzed (see Section 4 for how they were selected out of 310 response), 93 "realistic" projects were proposed in response to item 10, of which 70 proposals were elaborated enough to be mapped with an idealized approach

Table 3 Indications on which stakeholders are in position to positively impact STT

$N=310$	UMTs	School curriculum or reform committees	School teachers	Representatives of university students	Mathematics education researchers	Designers of technological resources
Strongly agree	42%	32%	46%	8%	20%	14%
Disagree or strongly disagree	2%	12%	2%	29%	22%	20%

for improving STT. Thirty-one projects (42% of 70) were restricted to the school level; 25 projects (34%) were restricted to the tertiary level; 7 projects (10% of 70) included changes both at the school level and at the tertiary level; and 7 (10% of 70) projects proposed some form of cross-community collaboration. The proposed projects were varied and represented all four ideal approaches for improving STT. Moreover, item 10 was the only item in the questionnaire that drew a considerable number of responses aligned with all four approaches. Here are a few examples:

> A preparation course just before the start of the university studies, led by university staff which is well-aware of the 'current' knowledge of the students and of the skills that will be required in the university, can be very helpful. (Jump-oriented)
> Some 'taste of university math' courses […] academics could visit and try to give an idea of university math (not some kind of 'popular math talk', but proof, formal reasoning and ideas from foundations). We already send to incoming students some info on what they'll be doing, but this doesn't seem to have much impact: face to face experience might. (Enculturation-oriented)
> University mathematics departments should interview many incoming STEM students to understand why they are pursuing their math-intensive STEM degrees and what excited them about these degrees. Then, the departments can better tailor our academic approach to meet these students' needs. (DEI-Oriented)
> School-funded regional professional development workshops for school teachers and university faculty (including [graduate teacher assistants], non-tenure-track and tenure-track faculty), including a panel of current college students. This way, all of these people can meet in person and think about issues together as well as vertical alignment issues. The panel of students would help to elaborate on their experiences. (Cross-community-oriented)

6 Discussion

To the best of our knowledge, this is the first study of *profound knowledge* (Lewis, 2015) on secondary-tertiary transition (STT) held by university mathematics teachers (UMTs), including their knowledge of desirable future measures for improving STT as rooted in the knowledge of past and present experiences. To recall, we see our study as an extension of Thomas et al.'s (2015) work, in which, in terms of the improvement science, the investigation focused mainly on *basic knowledge* of STT. Thomas et al.'s (2015) work indicated that over the last decade, the international mathematics education community has developed a substantial shared body of basic knowledge of students' difficulties in a few selected mathematical areas such as abstract algebra, calculus and analysis, and proof and proving. At the same time, Thomas et al.'s work highlighted a lack of basic knowledge of STT in other areas such as discrete mathematics. While this gap in basic knowledge of STT certainly needs to be filled, Thomas and colleagues concluded their work by calling also for research that can support coordination and dialogue within and across educational levels, which they found to be largely absent in the field. Accordingly, our aim in this study was to lay the foundations for development of a shared body of profound knowledge of STT, which in turn will support such coordination and dialogue.

Our study addressed three research questions which were aligned with fundamental questions of improvement science (Lewis, 2015) in the context of STT. A somewhat disturbing result related to the first question—what are UMTs looking to improve with respect

to STT if at all—is that most of the UMTs who took part in the survey see STT as an aggravating matter of concern in spite of greater awareness of STT and of widespread efforts to improve the situation. If 10–20 years ago UMTs' concerns regarding STT could have been attributed to lack of interest, accountability, or knowledge regarding improving STT, it seems not to be the case today. Indeed, 79% of the 310 participants in the survey indicated that at least two STT-related measures are implemented in their departments, and this is notably more than the already impressive results of Thomas et al.'s (2015) study. However, it appears that for most UMTs, this prevalence of measures does not translate to a sense that the situation is improving. This tacit sentiment—we do a lot, but the results are worse than expected and even than they were 10 years ago—can partially explain why a sizable part of the expressed perspectives on STT in our study as well as in Thomas et al.'s (2015) contained deficit narratives and were aligned with a jump-oriented approach on STT. Nevertheless, we find some encouragement in the fairly large pool of responses to various items in the survey that were aligned with enculturation-oriented, cross-community-oriented, and DEI-oriented approaches for improving STT, which suggest that the discourse on STT in mathematics departments is probably more refined and nuanced than Thomas et al.'s (2015) work implied.

In relation to the second question—how do UMTs determine whether there is an improvement—we briefly stop on the following result: measures aligned with the enculturation-oriented approach to dealing with STT, such as mentoring or tutoring, are generally considered the most effective. The more traditional (jump-oriented) measures, such as bridging courses, are still widespread, which is aligned with Thomas et al.'s findings, but, again, the discourse on these measures seems to be nuanced in the sense of aligning with more than one perspective on improving STT. We also note the host of criteria UMTs used to evaluate the effectiveness of measures for improving STT and the substantial variation in their estimation of the effectiveness of different measures. This variation is noteworthy considering that UMTs seem to generally agree about students' difficulties that these measures attempt to address, which suggests that this variation may have more to do with profound knowledge than basic knowledge.

Findings related to the third research question—what changes do UMTs propose for improving STT—elaborate an important and to our knowledge previously undocumented aspect of the UMTs' knowledge of STT. By and large, UMTs seem to be more in favor, in the context of future actions, of different perspectives on STT and improving it compared to their assertions about STT in the past and in the present. In particular, while we found no indication of ongoing communication on STT across institutions and education levels, which is consistant with Thomas et al.'s (2015) findings, we are encouraged by the strong presence of calls for tertiary-secondary collaborations and consider this to be a particularly promising avenue for further action. This is for both theoretical and practical reasons. First, such proposals are well aligned with one of the key principles of improvement science in education formulated by Bryk et al. (2015), namely, accelerating learning through networked communities. Second, calls for collaboration imply that UMTs can initiate or be receptive to initiatives taking a cross-community approach, which is an optimistic message to all the stakeholders involved in efforts to improve STT.

We conclude by revisiting the diversity of perspectives on improving STT. Comparison of our findings with Thomas et al.'s (2015) findings indicates that profound knowledge of STT may be much more diverse and fragmented compared to basic knowledge of STT. While UMTs and mathematics education researchers may generally agree about students' difficulties in different mathematical areas during STT, our study suggests that the notions "transition" and "improvement" have many different interpretations in the field. In this

study, we are not in a position to make any reliable conclusions about how profound knowledge of STT varies across contexts and communities, yet one may surmise that variation exists. Additional research of profound knowledge of STT is needed, both at the national and the international levels, towards improving communication on STT within and across educational levels, thus improving STT.

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Data availability The data that support the findings of this study is not publicly available due to privacy restrictions. Data are available from the corresponding author upon reasonable request and with the permission of the EMS Education Committee.

Declarations

Conflict of interest The authors declare no competing interests.

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